

Electronics Optocoupler

Optocoupler Operation as Switch Tutorials with Design Sample

by [electronicsbeliever](#) • 1 • Electronics, Optocoupler

The optocoupler operation as switch is similar on how to configure BJT as switch. For a BJT to operate as switch it must be set to function in saturation and cut-off. Optocoupler as well. However in BJT circuit the base current must be high enough to drive the transistor into saturation; but in Optocoupler it is the forward current. In BJT the base and collector currents are link to each other by the current gain of the transistor or simply beta. In Optocoupler circuit, the forward and the collector current are link to each other with the current transfer ratio or simply CTR.

$$\text{CTR} = (I_c / I_f) \times 100\%$$

To set optocoupler operation as switch; it must be driven into saturation. To saturate, the forward current must be big enough compared to the collector current. But is this possible? The collector current is always a function of the forward current and the device CTR.

Going back to BJT circuit; at saturation the transistor is no longer gaining thus the dependency of the collector current to the base current is no longer valid. It is the same scenario in Optocoupler. At saturation the circuit collector current is no longer link to the forward current through the device CTR. So in determining the collector current at saturation the above equation is no longer valid. Take note on this.

Optocoupler operation as switch is can be checked or set correctly by comparing the circuit CTR and the device CTR (CTR method). Circuit CTR is the ratio of the circuit collector to forward current.

$$\text{CTR}_{\text{circuit}} = (I_c \text{ circuit} / I_f \text{ circuit}) \times 100\%$$

The device CTR is the current transfer ratio of the device specified in the datasheet. At linear operation the circuit CTR is always following the device CTR. However, at saturation, the CTR demand of the circuit is very low compared to the CTR that can be provided by the device; thus the circuit will saturate. Just like you are asking a glass of water but someone gives you a jar of it; you cannot intake all of it.

Therefore at saturation the circuit CTR must be less than the device CTR.

$$\text{CTR circuit} < \text{CTR device}$$

Where

$$\text{CTR circuit} = I_c \text{ circuit} / I_f \text{ circuit}$$

Another way of checking an Optocoupler operation as switch is by computing the collector-emitter voltage (VCE method). At saturation its level is always zero or equal to the device VCEsat.

$$\text{VCE circuit} < = \text{VCEsat device}$$

Therefore collector-emitter voltage below this level is making the optocoupler operates in saturation. In this technique you can still use the device CTR. Do not be shocked if you can get a negative VCE level despite the circuit supply has no negative value. It's only implying that the device is operating at saturation. Then compute for actual collector current without using the device CTR value.

Design Example

Verify the circuit below if it can operate as a switch. Vdd and Vcc are both 5V. Consider 80% device CTR and the forward voltage of the diode is 1V.

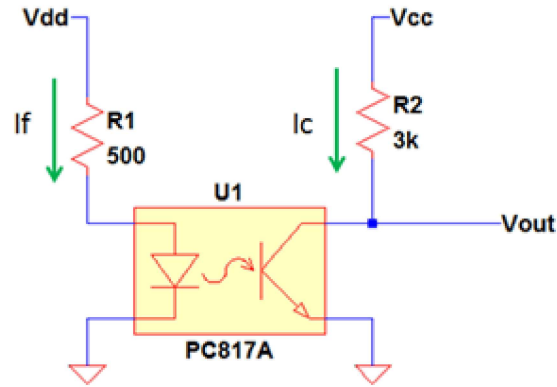


Figure 1

For Optocoupler operation as switch the collector-emitter voltage must be zero or just equal to the maximum saturation voltage specified in the datasheet. So in above circuit the node Vout must be zero since no specified VCEsat value.

Proving Optocoupler Operation as Switch using CTR Method

Solving for the forward current

KVL at loop 1

$$I_f = (V_{dd} - V_f) / R_1 = (5V - 1V) / 500 = 8 \text{ mA}$$

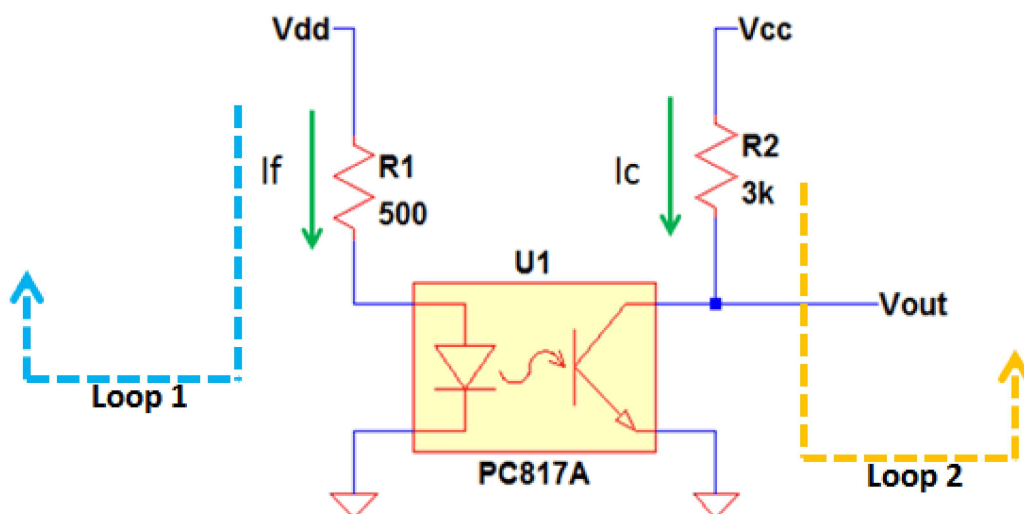


Figure 2

Solving for the collector current

The maximum collector current is

$$I_c = V_{cc} / R_2 = 5V / 3k = \mathbf{1.67\ mA}$$

Solve for the circuit CTR

$$CTR_{circuit} = I_c / I_f = 1.67\ mA / 8\ mA = \mathbf{20.875\%}$$

Comparing circuit CTR to device CTR

$CTR_{circuit}$ is less than CTR_{device} . No doubt the Optocoupler will saturate.

$$CTR_{circuit} < CTR_{device} = \mathbf{20.875\% < 80\%}$$

Proving Optocoupler Operation as Switch using VCE Method

No need to solve for the forward current, it's the same with above result.

Solving for the collector current

$$I_c = CTR_{device} \times I_f = 80\% \times 8\ mA = \mathbf{6.4\ mA}$$

Solving for collector-emitter voltage

From loop 2

$$V_{cc} - I_c \times R_2 - V_{CE} = 0$$

$$V_{CE} = V_{cc} - I_c \times R_2 = 5V - (6.4\ mA \times 3k) = \mathbf{-14.2\ V}$$

The computed VCE is -14.2V. How come it becomes negative that the supply has no negative? Do not be confused, the negative sign means that the Optocoupler is at saturation. This is also indicating that the device CTR is no longer valid in solving for the collector current. The true collector current is only dependent to the collector resistor.

TAGGED [HOW TO DESIGN AN OPTOCOUPLER SWITCH](#) [OPTOCOUPLER OPERATION](#)
[OPTOCOUPLER SWITCH](#)

[electronicsbeliever](#)

[More by electronicsbeliever](#)

Previous Article

[How to Determine Optocoupler
Operation Saturation or Linear](#)

Next Article

[Optocoupler Device CTR and Circuit
CTR Explained](#)

One thought on “Optocoupler Operation as Switch Tutorials with Design Sample”

Daniel Khadivi March 27, 2019 at 12:47 AM

Thank you for this great article and your other ones on optocouplers. I learned a lot about this great device and the important values in the datasheet and the graphs. The information you are sharing is very valuable. Practical quality material is hard to find. Most articles on the internet are very brief and devoid of any in-depth information. Electronic books are extremely impractical and boring. I am an electrical engineer by education,

many years ago, but never learned practical useful information in those books to build fun educative circuits to tinker with.

Please keep up your great work and thank you again for sharing your experience with the world.

REPLY

Follow Us



Electronics Believer
2,470 followers



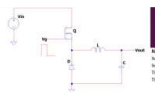
Quick Facts About Synchronous Buck Converter

The optocoupler operation as switch is similar on how to configure BJT as switch. For a BJT to operate as ... [Read more](#)

SUPPLY VOLTAGE			
Parameters	Value	Unit	
VDD	5	volt	This is the typical value of VDD. See the figure.
VDD	5	volt	This is the typical value of VDD. See the figure.
AMBIENT TEMPERATURE			
Parameters	Value	Unit	
Tamb max	60	°C	This is the maximum operating ambient temperature. This is a design given.
Tamb min	-30	°C	This is the minimum operating ambient temperature. This is a design given.
RESISTOR DATA			
Parameters	Value	Unit	
RG1	1500	ohms	Input here the resistance of RG1. See figure.
RG2	330000	ohms	Input here the resistance of RG2. See figure.
ps RG1 rating	0.1	watt	Input here the power dissipation rating of RG1. Get this from the datasheet.
ps RG2 rating	0.1	watt	Input here the power dissipation rating of RG2. Get this from the datasheet.
RELAY DATA			
Parameters	Value	Unit	
RCOIL_typical	100	ohms	Input here the coil resistance of the relay. Get this from the datasheet.
LCOIL	100	uH	Input here the coil inductance of the relay. Get this from the datasheet.
ps_coil rating	1	watt	Input here the power dissipation rating of the relay coil. Get this from the datasheet.
Tmin_relay	-40	°C	Input here the minimum operating temperature of the relay. Get this from the datasheet.
Tmax_relay	125	°C	Input here the maximum operating temperature of the relay. Get this from the datasheet.

Design Template for NMOS Lowside Relay Driver

The optocoupler operation as switch is similar on how to configure BJT as switch. For a BJT to operate as ... [Read more](#)

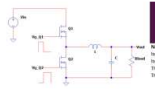


Buck Converter Design Template

Notes:
 Items in green are inputs. The user must enter values into these items.
 Items in orange are calculation outputs with formulas. The user should not alter them.
 This template is applicable to CCM and DCM modes (see the DCM mode equation).
 This is not applicable to synchronous buck.

Buck Converter Design Using Excel Calculator

The optocoupler operation as switch is similar on how to configure BJT as switch. For a BJT to operate as ... [Read more](#)



Synchronous Buck Converter Design Template

Notes:
 Items in green are inputs. The user must enter values into these items.
 Items in orange are calculation outputs with formulas. The user should not alter them.
 This template is applicable to CCM and DCM modes (see the DCM mode equation).
 This is not applicable to synchronous buck.

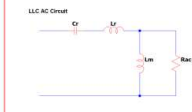
Synchronous Buck Converter Design Calculator Excel-External Switches

The optocoupler operation as switch is similar on how to configure BJT as switch. For a BJT to operate as ... [Read more](#)

Parameter	Symbol	Value	Unit
Input Voltage			
Minimum input voltage	V_{in_min}	100	V
Maximum input voltage	V_{in_max}	300	V
Maximum input voltage	V_{in_max}	450	V
Output Voltage			
Minimum output voltage	V_{out_min}	10	V
Maximum output voltage	V_{out_max}	15	V
Maximum output voltage	V_{out_max}	15	V
Load Current			
Minimum output voltage current (full load)	I_{out_min}	24	A
Maximum output voltage current (full load)	I_{out_max}	24	A
Maximum output voltage current (full load)	I_{out_max}	24	A
Frequency Targets			
Resonant Frequency	F_r (Resonant Frequency)	100	kHz
Switching Frequency	F_s (Switching Frequency)	400	kHz
Output Power			
Minimum output power	P_{out_min}	500	W
Maximum output power	P_{out_max}	500	W
Maximum output power	P_{out_max}	500	W
Inductor Value			
Inductor value (mH)	L (mH)	15	mH
Inductor value (mH)	L (mH)	15	mH
Capacitor Value			
Capacitor value (uF)	C (uF)	10	uF
Capacitor value (uF)	C (uF)	10	uF
Gate			
Minimum gate	V_{gs_min}	10	V
Maximum gate	V_{gs_max}	15	V
Maximum gate	V_{gs_max}	15	V

Template Legend

Input voltage: User must provide.
 Calculation result: With formula, do not alter.
 Calculation result: With formula, do not alter.
 Calculation result: With formula, do not alter.
 Calculation result: With formula, do not alter.



Half Bridge LLC Resonant Converter Power Section Calculator

The optocoupler operation as switch is similar on how to configure BJT as switch. For a BJT to operate as ... [Read more](#)

BJT (11)

Boost Converter (3)

Buck Converter (7)

Circuit Analysis (1)

Circuit Analysis (1)

Control System (1)

Design Template (5)

Diode (2)

Electronics (45)

EMI/EMC (1)

ESD (2)

EV Charging (1)

Flyback Converter (1)

Fuse (3)

How To (14)

Inverting Buck-Boost (1)

[LED Drivers](#) (1)
[LLC](#) (5)
[LTSpice](#) (9)
[Mathcad](#) (1)
[Matlab](#) (2)
[Miscellaneous](#) (3)
[MOSFET](#) (17)
[Optocoupler](#) (11)
[Others](#) (1)
[Power Electronics](#) (26)
[Power Supply](#) (6)
[Quasi Resonant Flyback](#) (9)
[RC Circuit](#) (1)
[Regulators](#) (2)
[Relay](#) (2)
[Reviews](#) (5)
[Simulink](#) (2)
[Thermal Management](#) (1)
[Troubleshooting](#) (3)
[Tutorials](#) (31)



